Operations Research - Project E

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1 Project E

Here I report the detailed description of how I solved the problem. In some sections, I explicitly described all the reasoning behind a chosen solution.

1.1 Analysis of the problem

The problem under study is the following.

> We want to build a skyscraper in the historic center of Padua. The workers who build the steel structures of skyscrapers are highly specialized, therefore difficult to find. The table below shows the requests for workers on site in a six-month period. Transfers from other construction sites in Padua are only possible on the first of each month and cost 100 euros per worker. At the end of each month, workers can go to other sites for 160 euros. Over/understaffing costs 200 euros per month per person (in case of lack of staff, the missing hours must be covered through the overtime of the available workers).
>
> | March | April | May | June | July | August |
> |---|---|---|---|---|---|
> | 4 | 6 | 7 | 4 | 6 | 2 |
>
> Overtime must be limited to 25% of the total hours worked. Every month, at most three workers can arrive at the construction site. The departure of workers to other construction sites is limited, due to the agreements with the trade unions, to one third of the total staff employed in the month. Suppose that 3 workers are already present on site at the end of February, that no one is transferred at the end of February and that 3 workers remain on site at the end of August. What is the number of departures and arrivals that minimizes the total cost?

Figure 1: The problem

1.1.1 Main features of the problem

From the problem we can extract some peculiar features of the problem.

Firstly, we observe that, in general, we need to satisfy at each month a given fixed demand, i.e., the numbers of workers. This makes us understand-

ing that it is a problem similar to the family of production problems with inventory management (PPIM).

A major difference from PPIM is that it is not strictly necessary to fulfill the demand at each month, i.e., we can partially satisfy the demand. We can imagine the overstaffing cost as a penalty to "stock" an extra worker and the understaffing cost as a penalty for a missing worker.

We have two production costs: the cost to transfer to Padua a new worker and the cost to transfer from Padua an employed worker.

Then we need to deal with other constraints as the overtime and the maximum number of arrivals and departures.

Obviously, we are treating an ILP since all the quantities are indivisible and we observe that the number of workers that will arrive will be equal to the number of workers that will leave.

1.1.2 Objective function

We can start by setting the first variables and parameters we need and define the objective function. Let T be the set of months and we can suppose that $T = \{1, 2, 3, 4, 5, 6\}$ by associating March with 1, April with 2 and so on and so forth.

For each period, we denote by x_t the number of workers that will arrive in month t and by y_t the number of workers that will leave in month t. The cost to transfers to Padua and from Padua is the same in each month. Hence, we denote those two, with p and k respectively.

Then we have to add the understaffing and overstaffing costs. For both, there is a fixed cost v (which also does not depend by the month). I decided to denote with u_t and o_t the number of workers in overstaffing and understaffing.

Hence, the objective function is

$$\min \quad \sum_{t=1}^{n} (px_t + ky_t + v(o_t + u_t)). \tag{1}$$

For the sake of explanation I'll show later why I decided to put $v(o_t + u_t)$ when dealing with the over/understaffing costs.

1.1.3 Conservation principle constraint

Somehow we need a constraint that assure that the number of the worker at the end of month t is equal to the number of workers that remained at the

end of month $t-1$ (i.e., those workers that remained at the end of month $t-1$) plus the remaining workers after all the transfers in month t.

I denote with h_t the workers available at the end of month t and set our first constraint

$$h_t = h_{t-1} + x_t - y_t, \quad t \in T, \tag{2}$$

where we also have $h_0 = 3$, which is the numbers of workers available at the end of February. Thus, we can set $h_6 = 3$, imposing that at the end August there will be 3 workers.

Observe that the employed workers in month t are given by $h_{t-1} + x_t$.

1.1.4 Overstaffing and understaffing constraints

Now, let's deal with the over/understaffing constraints and in particular with the overstaffing one since the other is similar. For this reason I'd like to show all the reasoning I did. At the beginning I thought at something in the form:

$$\begin{aligned} h_{t-1} + x_t - d_t &\leq C y_t, \\ h_{t-1} + x_t - d_t &= o_t, \end{aligned}$$

where C is an artificial variable (for example the sum of all of the d_t) and y_t takes value in $\{0, 1\}$. Then, putting in the objective function $o_t y_t$ times the cost for overstaffing (v). With this solution, whethever in month t, the number of workers is strictly greater than the demand, we have $y_t = 1$ and the first constraint becomes irrilevant and o_t is exactly the numbers of extra workers. Otherwise, we have $y_t = 0$ and in the objective function we have $o_t y_t = 0$.

Now we would have to properly replace $o_t y_t$ (since is the product of a binary variable and of an non-negative integer variable) but I came up with a better solution, which is:

$$h_{t-1} + x_t - d_t \leq o_t, \tag{3}$$

with $o_t \geq 0$. Hence, we can get rid off of y_t and just leaving $v \cdot o_t$ in the objective function. If in month t, the number of workers is strictly greater than the demand, o_t will be the exact number of extra workers (since the solver will choose the smallest integer to make valid the inequality). On the other hand, if $h_{t-1} + x_t - d_t \leq 0$, we'll have $o_t = 0$ and thus no overstaffing penalty.

Similarly for the understaffing constraint we have:

$$d_t - (x_t + h_{t-1}) \leq u_t, \tag{4}$$

with $u_t \geq 0$ and $v \cdot u_t$ in the objective function. If in month t, the number of workers is strictly less than the demand, u_t will be the exact number of missing workers. On the other hand, if $d_t - (x_t + h_{t-1}) \leq 0$ (there are more workers than the demand), we'll have $u_t = 0$ and thus no understaffing penalty.

Observe that we cannot have both o_t and u_t strictly greater than 0 at the same time. If $o_t > 0$ then $x_t + h_{t-1} - d_t > 0$ but this means that $d_t - (x_t + h_{t-1}) < 0$. Thus u_t will be 0. Then at most just one of the these two penalties is active.

1.1.5 Overtime constraint

Since there is a constraint about the overtime, the model cannot allow that the available workers sono troppo pochi. In particular, each worker cannot work a 25% more than expected. We can formulate this as

$$\frac{d_t}{x_t + h_{t-1}} \leq 1.25, \tag{5}$$

which is equivalent to

$$d_t \leq 1.25(x_t + h_{t-1}). \tag{6}$$

1.1.6 Other constraints

We have an upper bound to the maximum number of worker that can arrive in a certain month t which is given by

$$x_t \leq 3. \tag{7}$$

Finally, we need a constraint on the number of workers that can leave in a month t. We know that the departure of workers to other construction sites is limited, to one third of the total staff emploted in the month t. So,

$$y_t \leq \frac{h_{t-1} + x_t}{3}. \tag{8}$$

Observe that in this way we also ensure that the numbers of worker that leaves cannot be grater the total staff employed in month t.

Finally, we impose $x_t, y_t \geq 0$ and observe that we do not need $h_t \geq 0$ because of its definition.

1.2 The model

In conclusions, here I report the final model with the variables and the parameters.

1.2.1 Variables

The fixed parameters are:

- T: set of six periods (months)
- d_t: demand of workers in month t;
- p: cost to transfer a worker to Padua;
- k: cost to transfer a worker from Padua;
- v: penalty to pay in case of understaffing or overstaffing.

The variables of the model are the following.

- o_t: number of workers in overstaffing in month t;
- u_t: number of workers in understaffing in month t;
- x_t: number of workers that will arrive in month t;
- y_t: number of workers that will leave in month t;
- h_t: number of workers that will remain at the end of month t.

1.2.2 Model

The formulation of the problem is

$$\min \quad \sum_{t=1}^{n} (px_t + ky_t + v(o_t + u_t)) \tag{9}$$

s.t.

$$x_t \leq 3, \quad t \in T \tag{10}$$

$$h_t = h_{t-1} + x_t - y_t, \quad t \in T \tag{11}$$

$$y_t \leq \frac{h_{t-1} + x_t}{3}, \quad t \in T \tag{12}$$

$$x_t + h_{t-1} - d_t \leq o_t, \quad t \in T \tag{13}$$

$$d_t - x_t - h_{t-1} \leq u_t, \quad t \in T \tag{14}$$

$$d_t \leq 1.25 \cdot (x_t + h_{t-1}), \quad t \in T \tag{15}$$

$$h_6 = 3, h_0 = 3, \tag{16}$$

$$x_t, y_t, o_t, u_t \geq 0, \quad t \in T \tag{17}$$

1.3 Implementative details

The computational implementation of the model and its data can be found in *skyscrapE.mod*, *skyscrapE.dat* and *skyscrapE.run*. Here I discuss some implementative choices I have made.

1.3.1 Handling the initial condition and the set of the months

I decided to consider the months set T as circular queue containing only the months from March to August. Then I defined a new parameter h_init, initialized with the value of the number of available workers at the end of February (3). Finally a new symbolic parameter $first_month$ which tells what is the starting month to consider.

In this way I could easily handle, for the constraints (11)-(15), the h_0 variable. By means of an *if-else* condition, I imposed that when $t = 1$ (=$first_month$), the value of h_{t-1} must be retrieved from h_init, otherwise from the vector h.

1.3.2 Choice of the solver

Since the problem is an ILP and it's small and not too complex, I chose the *cplex* solver.

1.4 Optimal solution of the problem

The optimal value of the minimization problem is $f = 1780$€. The optimal solution is given in the table below.

	x_t	y_t	h_t	o_t	u_t
March	1	0	4	0	0
April	2	0	6	0	0
May	0	1	5	0	1
June	0	0	5	1	0
July	0	1	4	0	1
August	0	1	3	2	0

Table 1: Final value in each variable.

We have three arrivals: one in March e two in April; and three departures: one in May, in July and in August. The total number of arrivals and departures is six. As expected, the number of arrivals is equal to the number of departures.